Responsible AI: Algorithmic fairness and explainability

Introduction to the course and Fairness 1
3.9.2026

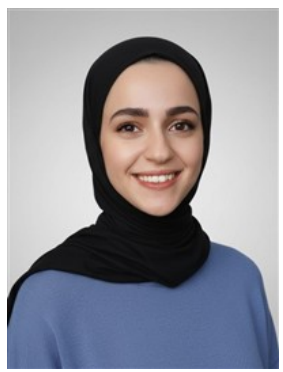
Aasa Feragen
afhar@dtu.dk

Welcome to Responsible AI: Algorithmic fairness and explainability

We really look forward to working with you!

- **Teachers:** Stella Frank (stefra@dtu.dk) and Aasa Feragen (afhar@dtu.dk, course responsible)
- **TAs:** Aya El Gebaly (aafel@dtu.dk) and Aditya Parikh (adipa@dtu.dk)






We are excited to work with you!

Course structure

- We meet every Thursday
- Lecture 8:30-10:00, exercises/discussions 10:00-12:00
- The course has 4 parts:
  - Algorithmic fairness for predictive models
  - Algorithmic fairness for generative models
  - Explainable AI for predictive models
  - Explainable AI for generative models
- Additionally, we have a strong focus on paradigms and limitations of machine learning
- Each part has a group project, summarized in a poster
- Let's look at the schedule!

Exam

- Projects are pass/fail; you need to pass all 4 projects to take the exam
- There will be a single resubmission possibility at the end of the course where you will be able to resubmit one of the projects
- The exam will either oral or a multiple choice written exam, to be decided within the next month – regardless, will cover both projects and entire curriculum
- Questions?

This is an in-person course!

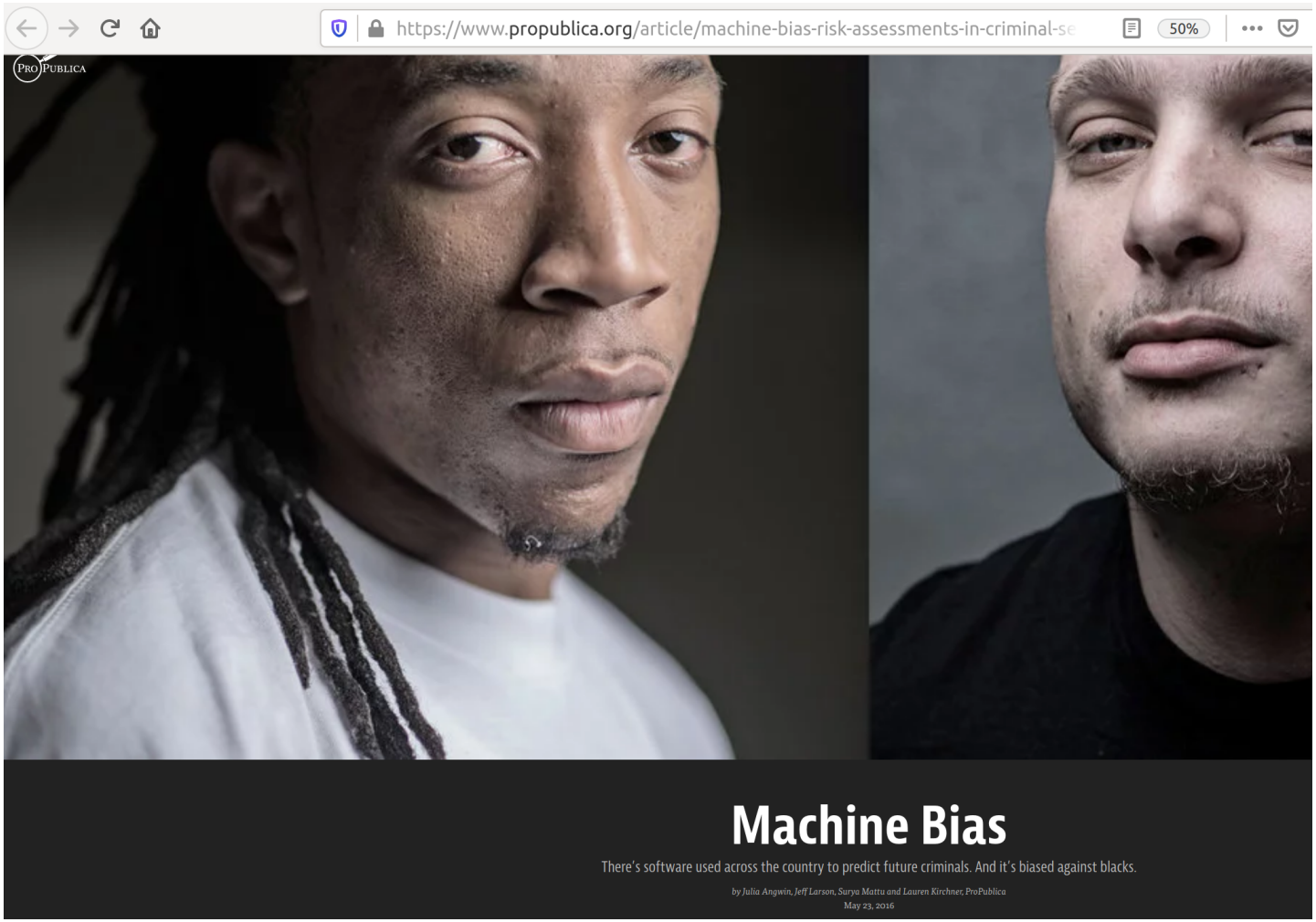
- Critical discussion and peer feedback are crucial elements of this course
- Group work are mandatory to take the exam

Use of AI in this course

- You are allowed to use AI as a tool during the projects
- **However**, consider what you want from your education:
  - If the AI does the work for you, you are just adding noise â the AI alone would do better
  - If you use the AI to help you learn, to get past hurdles, but still make sure to be the main actor in understanding and carrying out the project, you have a better chance of playing a crucial part
  - You will not have access to AI during the oral exam
- How do I know if I rely too much on the AI?
  - Try to carry out the exercise yourself, before getting AI help
  - Use the AI to find your errors, not to solve the exercises for you
  - Come to class, engage with your fellow students and teaching, and make sure you are up to date

Why Responsible AI?

A “classical” example: The COMPAS algorithm



Respectful discussions

Learning goals: Algorithmic fairness 1

By the end of today's lecture, you should:

- Be familiar with different ways in which demographic bias can become embedded in machine learning models
- Be familiar with the three most common fairness criteria, and variants thereof
- Know how to turn fairness criteria into diagnostic criteria for algorithmic bias
- Be familiar with incompatibility results for fairness criteria
- Be ready to start playing with algorithmic bias detection and mitigation

Introduction to Algorithmic Fairness

Fairness definitions based on the classification chapter of `https://fairmlbook.org/`

Fairness in AI

Google apologizes after Photos app tags black couple as gorillas: Fault in image recognition software mislabeled picture

- New York based computer programmer spotted images of himself and a friend had been labelled in an album marked 'gorillas' by Google Photos
- Google's image recognition software had automatically grouped the photos
- Google has said it is 'appalled' and 'genuinely sorry' for the mistake
- The company is working on ways to prevent similar errors in the future

By RICHARD GRAY FOR MAILONLINE
PUBLISHED: 13:39 GMT, 1 July 2015 | UPDATED: 21:17 GMT, 1 July 2015

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Google has been forced to apologise after its image recognition software mislabelled photographs of black people as gorillas.

The internet giant's new Google Photos application uses an auto-tagging feature to help organise images uploaded to the service and make searching easier.

However the software has outraged users after it mislabelled images of a computer programmer and his friend as the great apes.

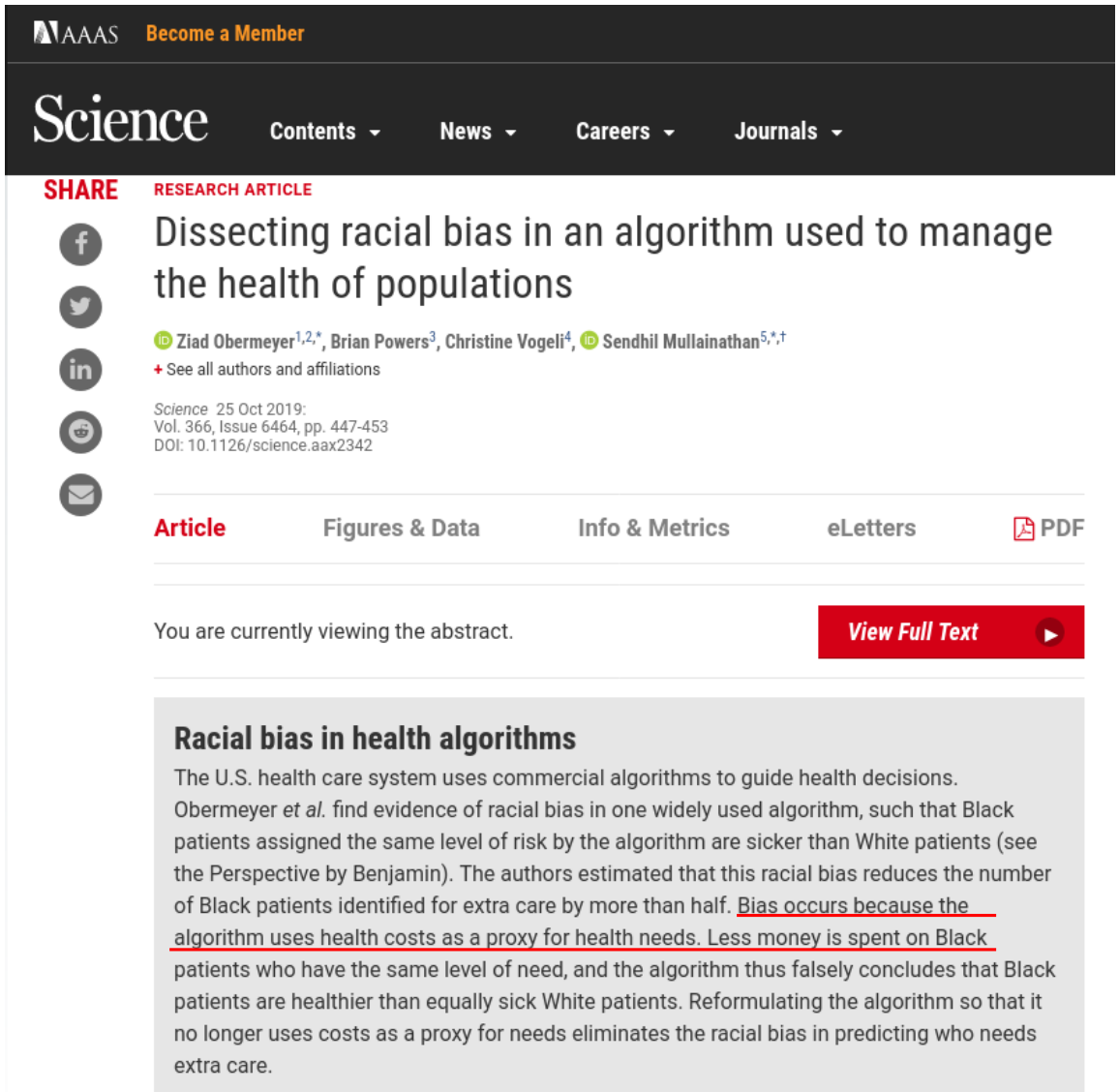


Google has issued an apology after computer programmer Jacky Alcine, from New York, spotted photographs of himand a femalefriend had been labelled as gorillas by Google Photos image recognition software. He sent aseries ofTweets to Google highlighting theproblem (like above) leading Google to issuea fix for the problem

Google said it was 'appalled' and 'genuinely sorry' for the mistake.

The fault comes just over a month after Flickr's autotagging system placed potentially offensive tags on images including mislabelling concentration camps as 'jungle gyms' and people as apes.

Fairness in AI



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RESEARCH ARTICLE

Dissecting racial bias in an algorithm used to manage the health of populations

Ziad Obermeyer[1,2,*], Brian Powers[3], Christine Vogeli[4], Sendhil Mullainathan[5,*,†]

+ See all authors and affiliations

Science 25 Oct 2019:
Vol. 366, Issue 6464, pp. 447-453
DOI: 10.1126/science.aax2342

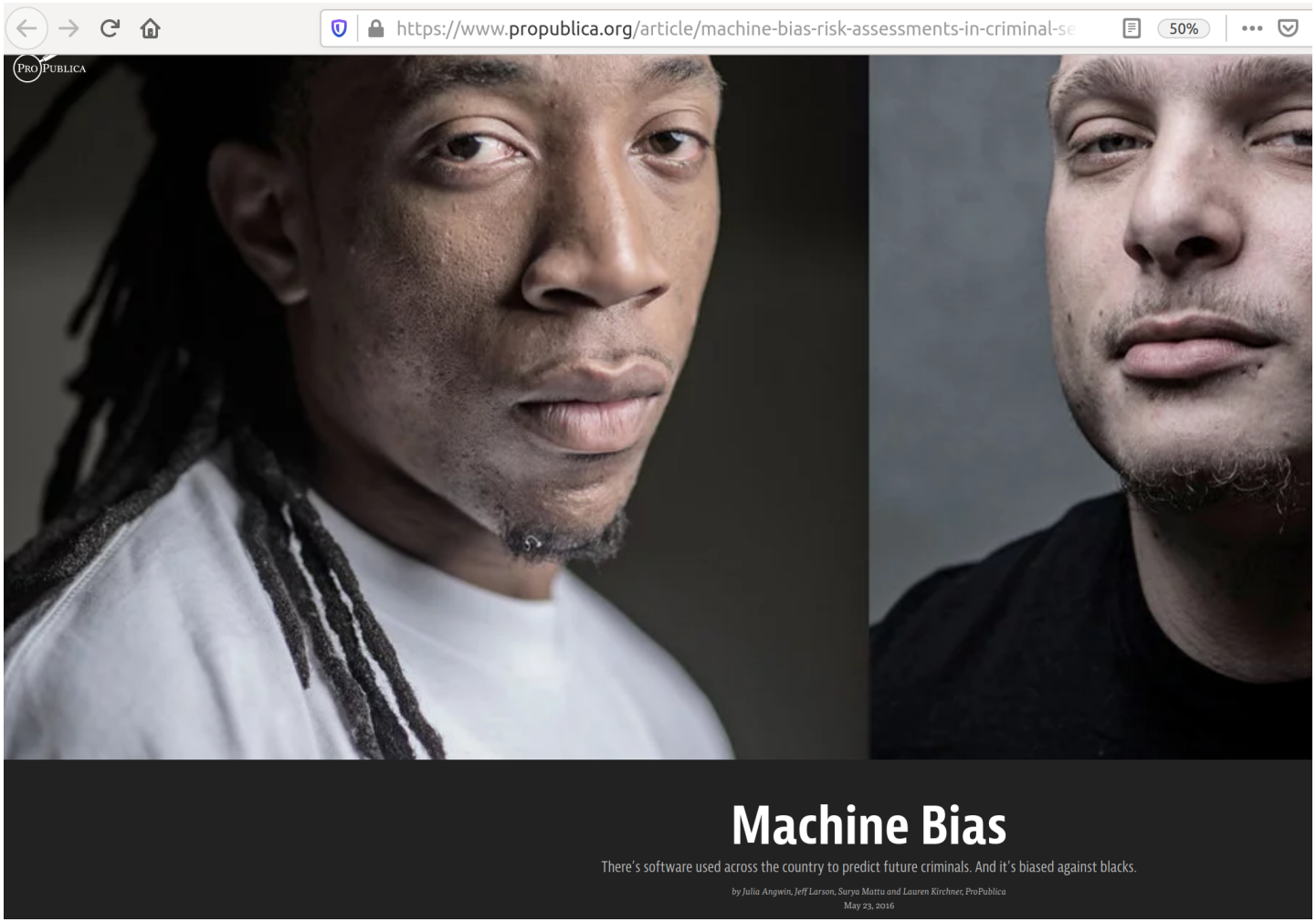
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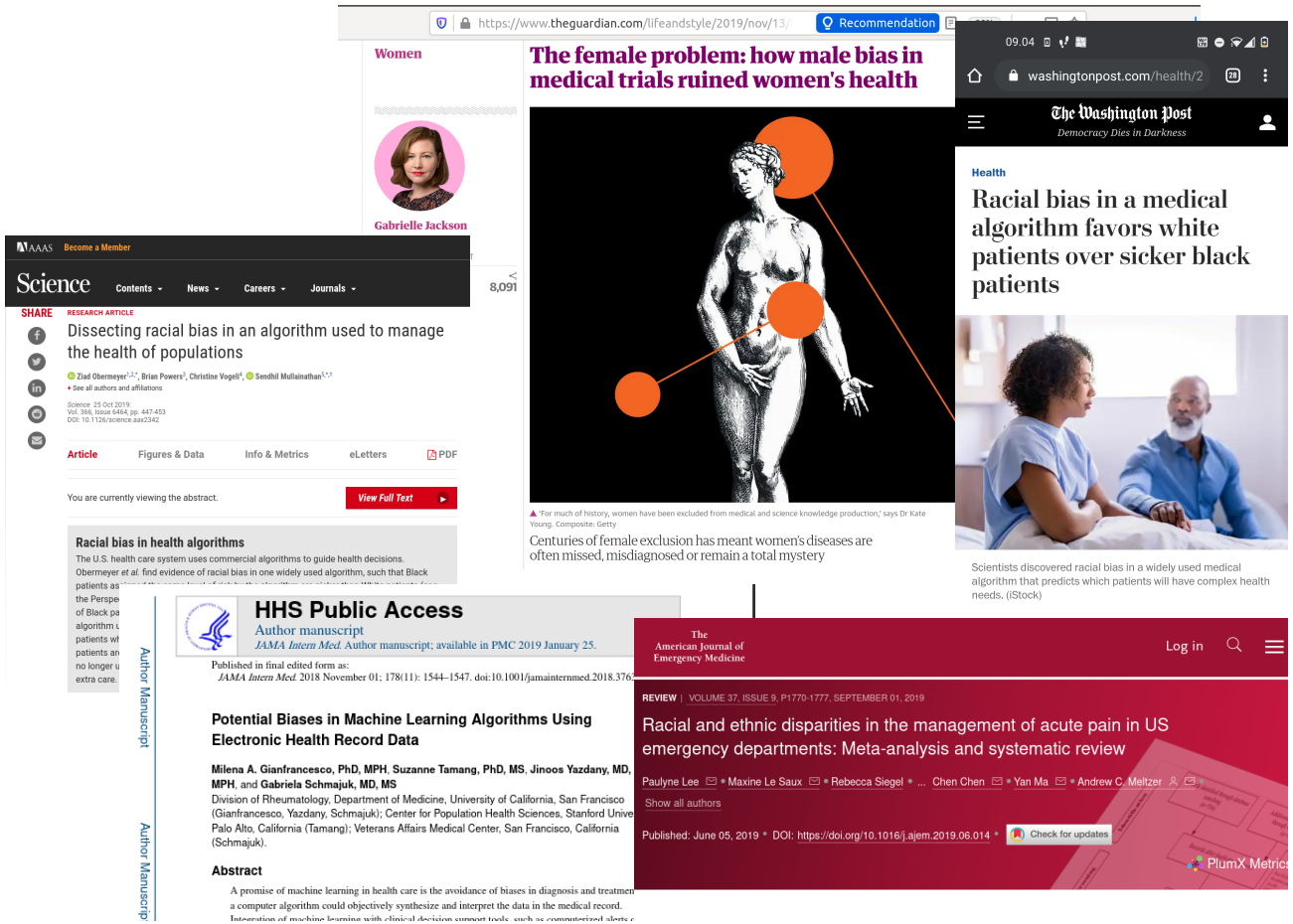
Racial bias in health algorithms

The U.S. health care system uses commercial algorithms to guide health decisions. Obermeyer *et al.* find evidence of racial bias in one widely used algorithm, such that Black patients assigned the same level of risk by the algorithm are sicker than White patients (see the Perspective by Benjamin). The authors estimated that this racial bias reduces the number of Black patients identified for extra care by more than half. Bias occurs because the algorithm uses health costs as a proxy for health needs. Less money is spent on Black patients who have the same level of need, and the algorithm thus falsely concludes that Black patients are healthier than equally sick White patients. Reformulating the algorithm so that it no longer uses costs as a proxy for needs eliminates the racial bias in predicting who needs extra care.

Fairness in AI



Fairness in (healthcare) AI



A simulated example

Imagine using predicted depression risk scores for prioritizing resources such as referral to a psychologist

(It's not exactly hypothetical)

Forbes

FORBES > INNOVATION > AI

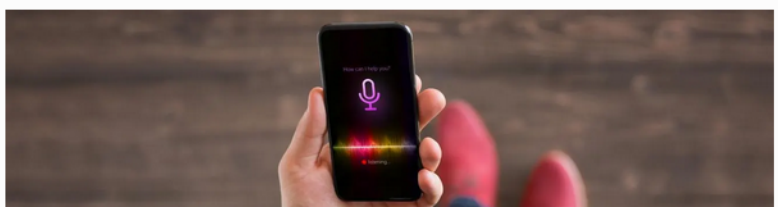
AI Can Now Detect Depression From Your Voice, And It's Twice As Accurate As Human Practitioners

Ganes Kesari Contributor
2X Founder & Chief Decision Scientist, TEDx Speaker, Adjunct Professor

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May 24, 2021, 06:50am EDT

This article is more than 2 years old.



Bias in algorithms: A toy illustration

It is well known that:

- Depression is diagnosed more frequently in women than in men
- This can partially be explained by different cultural perceptions of women and men (Sigmon et al, 2005)



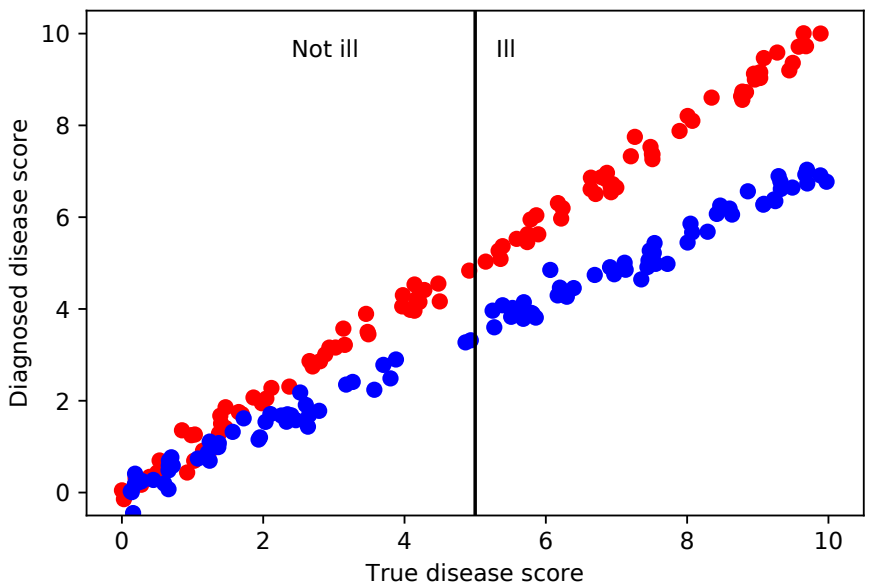
- If the diagnostic criteria are adapted to male symptoms, then the prevalence of depression among men increases (Martin et al, 2013)

If the data used for training ML algorithms to predict depression risk is skewed, then the trained algorithm will produce skewed predictions – it will be unfair. Let's simulate this.

Bias in algorithms: A toy illustration

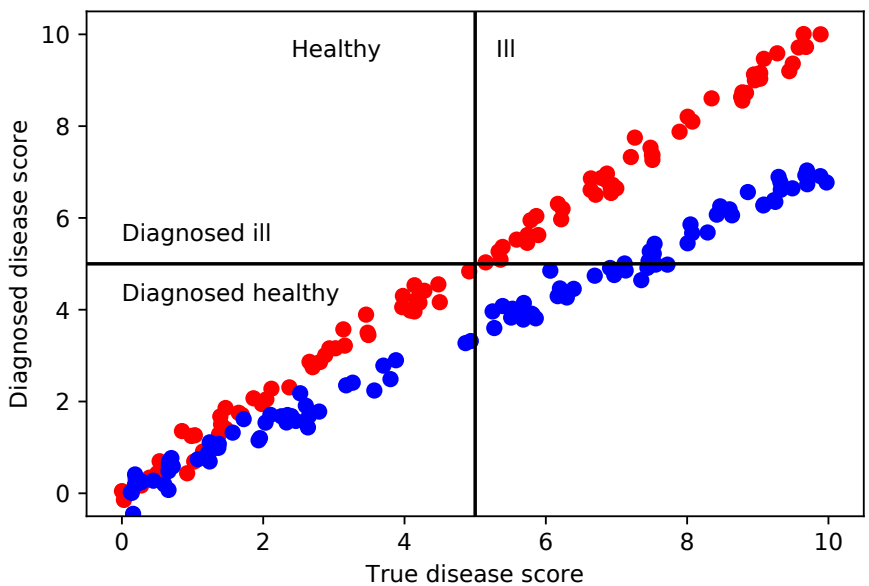
Imagine a disease model where

- Disease is scored from 0=healthy to 10=severe
- A true diagnosis corresponds to true score > 5
- Blue people (e.g. men) are systematically underdiagnosed due to differences in cultural perceptions of gender (e.g as with depression, Sigmon et al. 2005)



Bias in algorithms: A toy illustration

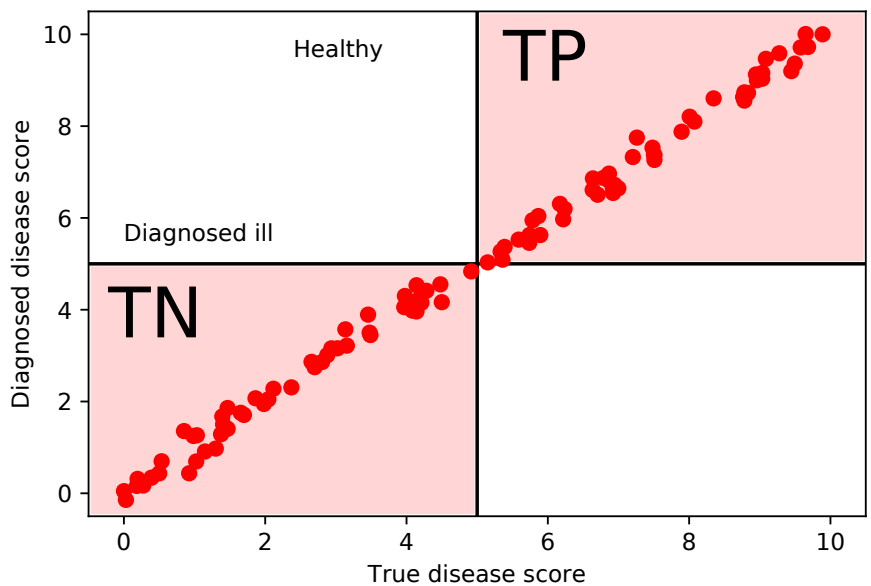
Setting a diagnostic threshold at diagnosed disease score = 5, we see that:



Bias in algorithms: A toy illustration

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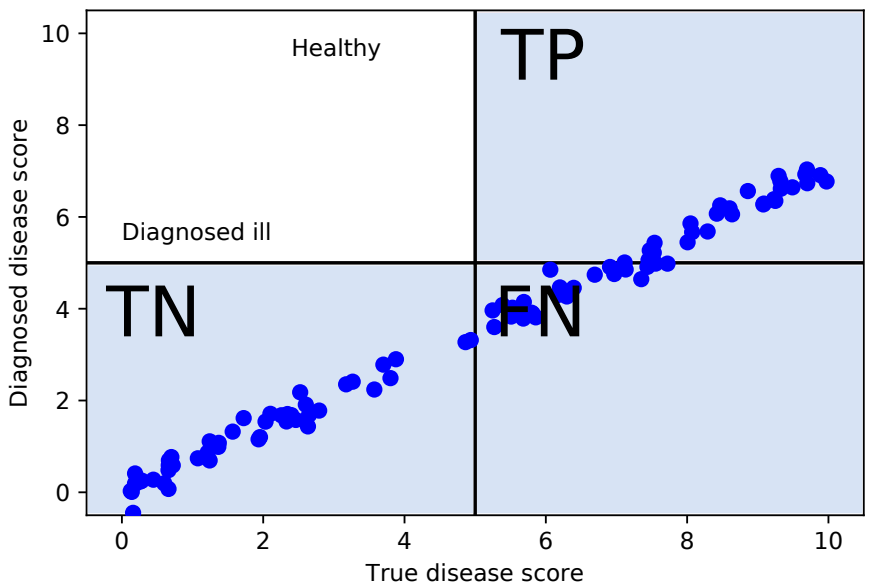
- For the red group, we have no false diagnoses



Bias in algorithms: A toy illustration

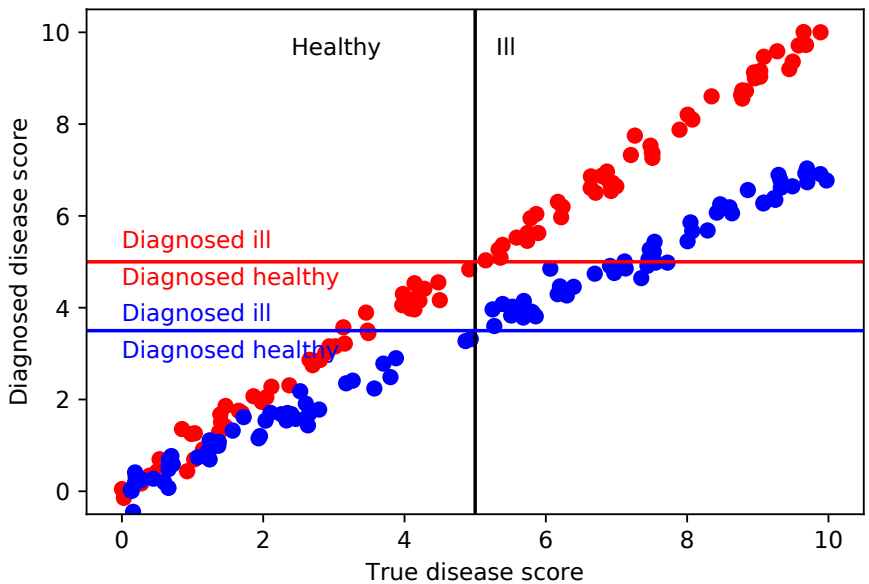
Setting a diagnostic threshold at diagnosed disease score = 5, we see that:

- For the red group, we have no false diagnoses
- For the blue group, false negative diagnoses are made



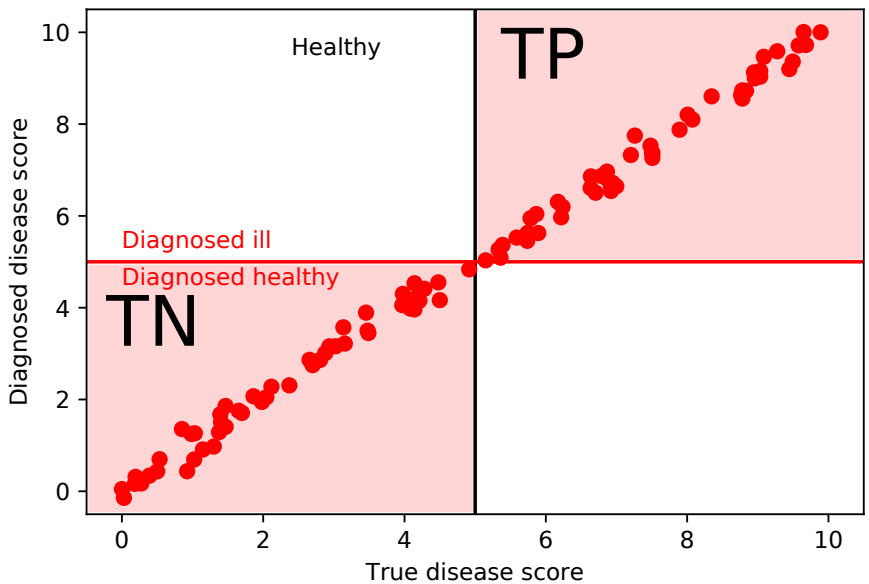
Bias in algorithms: A toy illustration

Solution: Population-specific thresholds



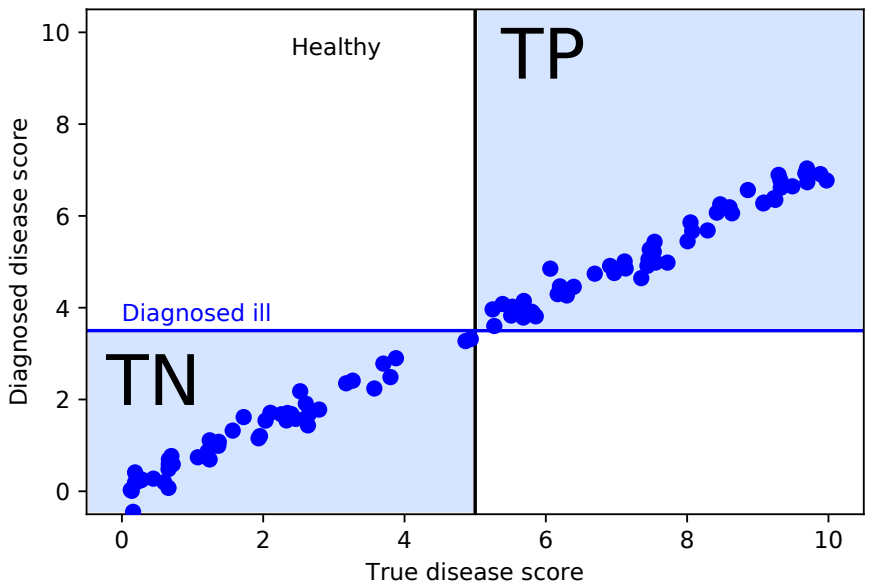
Bias in algorithms: A toy illustration

Solution: Population-specific thresholds



Bias in algorithms: A toy illustration

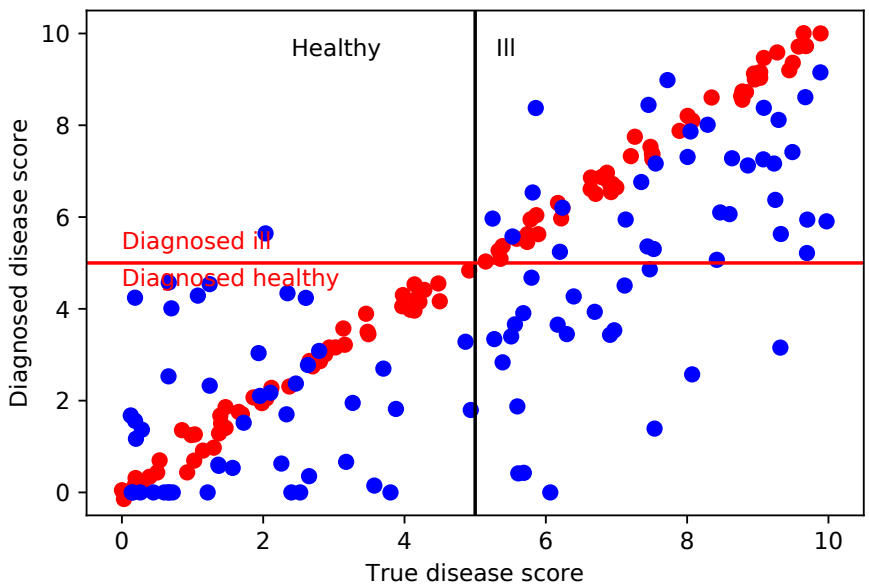
Solution: Population-specific thresholds



Bias in algorithms: A toy illustration

In a different disease model, the diagnostic criteria are more appropriate for the red group than for the blue, as in (Martin et al, 2013)

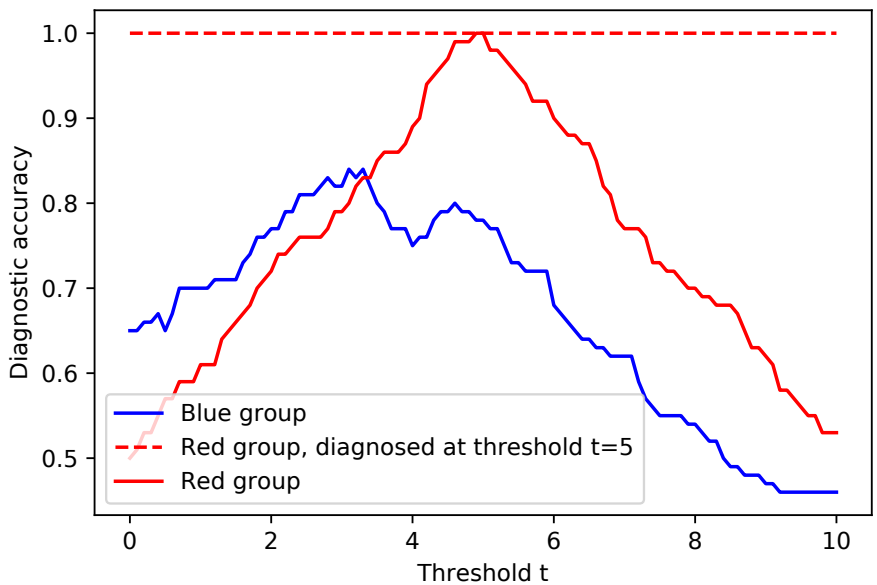
- Here, the score=5 threshold creates false positives and negatives in the blue group



Bias in algorithms: A toy illustration

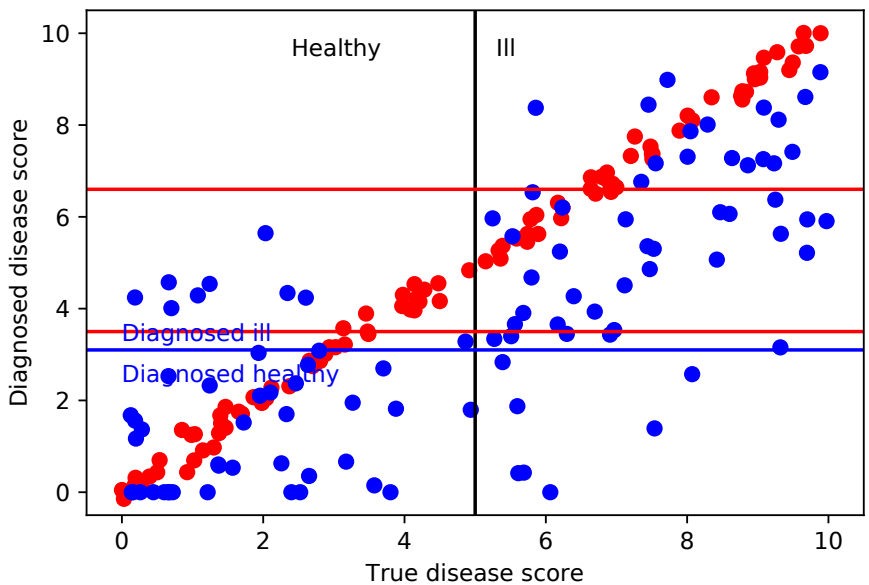
Below, see the group-wise diagnostic accuracy for the two different classes

- We are uncapable of reaching perfect accuracy for the blue group
- Two thresholds for the red group give the same accuracy as the best seen for the blue group



Bias in algorithms: A toy illustration

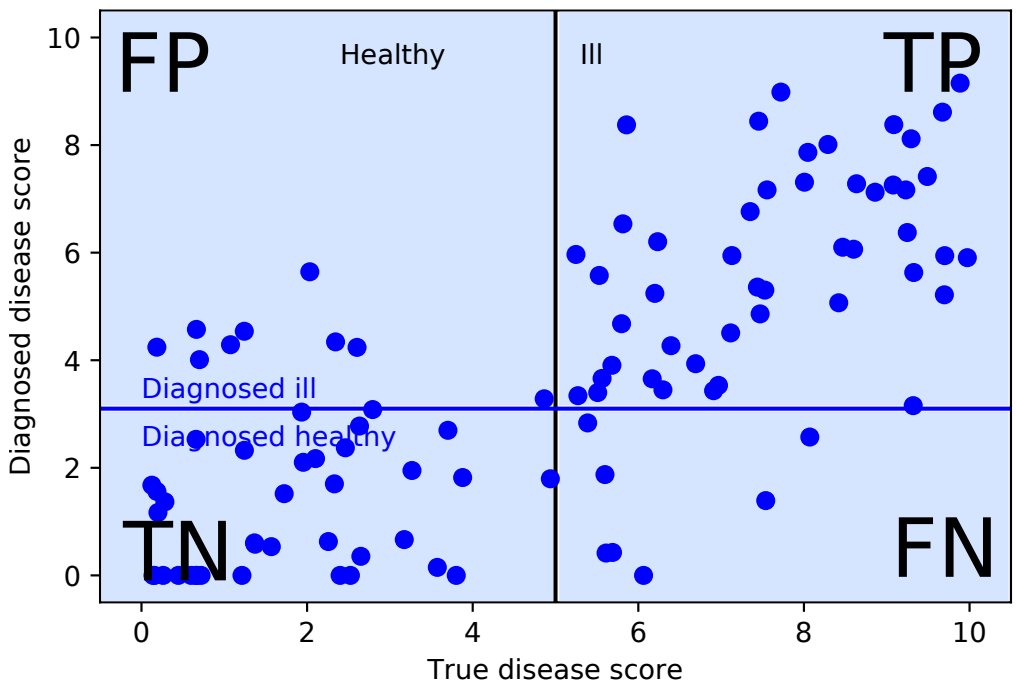
Let's see what those thresholds do:



Bias in algorithms: A toy illustration

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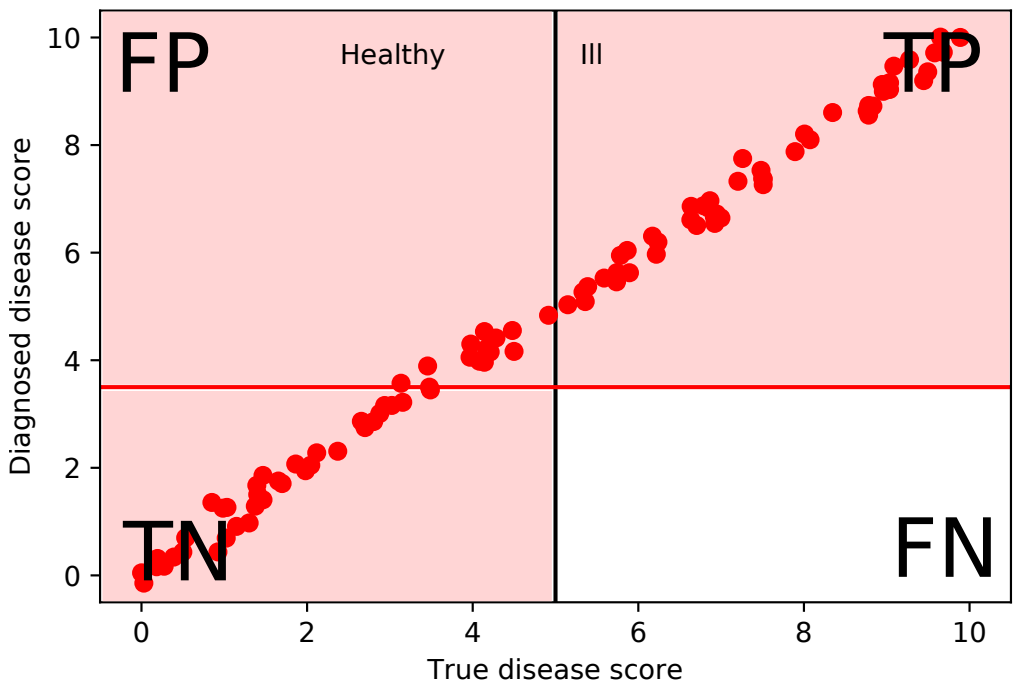
- Blue group has positive TP, TN, FP and FN



Bias in algorithms: A toy illustration

Let's see what those thresholds do:

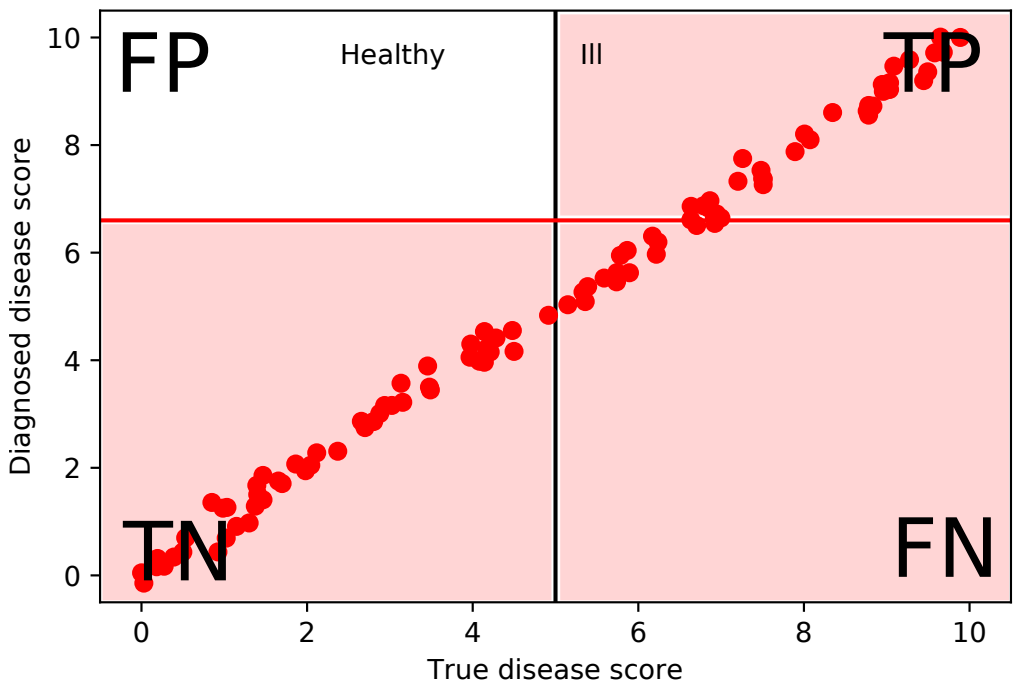
- Blue group has positive TP, TN, FP and FN
- Red group has positive TP, TN and FP, but no FN



Bias in algorithms: A toy illustration

Let's see what those thresholds do:

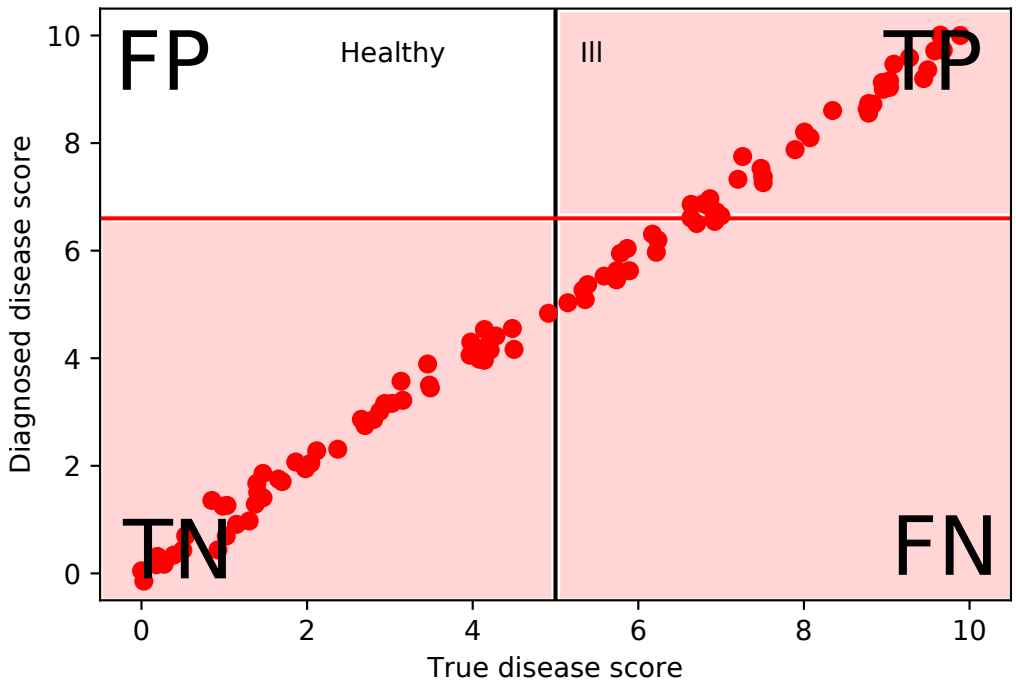
- Blue group has positive TP, TN, FP and FN
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Bias in algorithms: A toy illustration

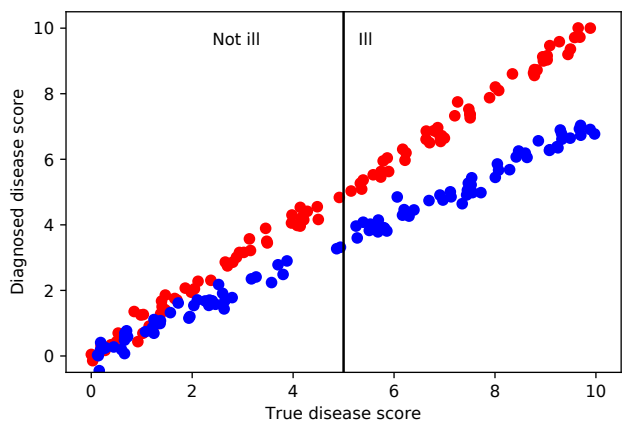
Let's see what those thresholds do:

- Blue group has positive TP, TN, FP and FN
- Red group has positive TP, TN and FN, but no FP
- **Note:** Although we have *sacrificed performance* in the red group, we still have a *bias* in our errors.



What *is* bias?

- Over- or under-representation is not necessarily a discriminating bias in itself – for instance, breast cancer *is* more prevalent in women than in men
- Data- and algorithmic bias usually refers to *systematic errors* that differ between groups.
- In order to detect this bias, we need to access the true labels (e.g. true diagnosis)
- This is often impossible – thus, our analysis depends on finding a reliable *proxy* for the true label.



Let's formalize this: Fairness criteria

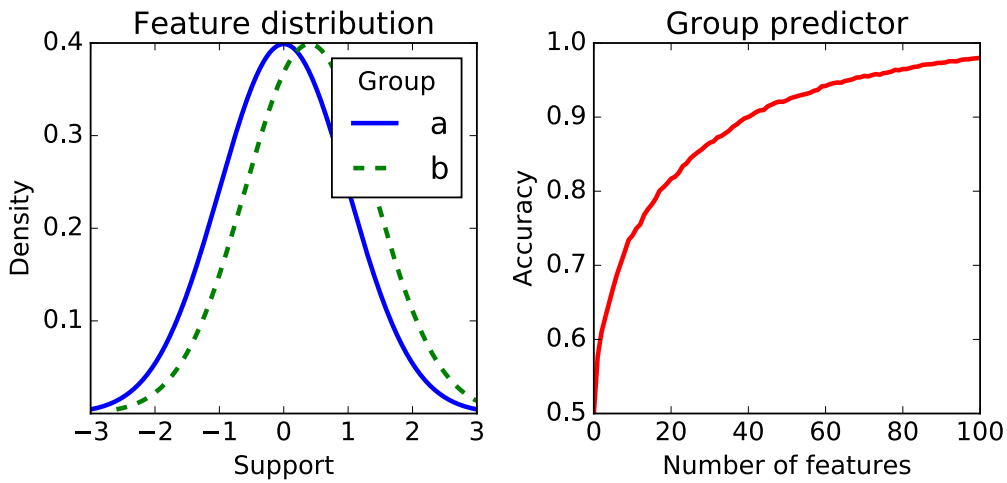
https://fairmlbook.org/

Classification: Fixing notation

- Input features (covariates) X, targets Y – we consider binary classification where $Y \in \{0, 1\}$
- Classifier (e.g. neural network) $f(X) \approx Y$
- Models generally predict a probabilistic *score* $f(X) = R$, which can be thresholded to obtain a binary $\hat{Y}$

Sensitive attributes - no fairness through unawareness

- Denote the set of sensitive groups by $A = \{a, b\}$
- No fairness through unawareness! Deleting the sensitive attribute will not solve the problem
- Below, consider features that are distributed slightly differently with respect to group – together, they encde the group



- To formalize fairness of classifiers, we study how the predictions $\hat{Y}$ and the model scores R that generated them depend on the sensitive groups in A.

Classification metrics

- Input features (covariates) X, targets Y – we consider binary classification where $Y \in \{0, 1\}$
- Classifier (e.g. neural network) $f(X) = R$ thresholded to give $\hat{Y}$ predicting Y

A range of different classification metrics can be used to define fairness:

Common classification criteria

Event	Condition	Resulting notion ($\mathbb{P}\{\text{event} \mid \text{condition}\}$)
$\hat{Y} = 1$	$Y = 1$	True positive rate, recall
$\hat{Y} = 0$	$Y = 1$	False negative rate
$\hat{Y} = 1$	$Y = 0$	False positive rate
$\hat{Y} = 0$	$Y = 0$	True negative rate

Additional classification criteria

Event	Condition	Resulting notion ($\mathbb{P}\{\text{event} \mid \text{condition}\}$)
$Y = 1$	$\hat{Y} = 1$	Positive predictive value, precision
$Y = 0$	$\hat{Y} = 0$	Negative predictive value

Different notions of algorithmic fairness

https://fairmlbook.org/

Three different categories of "fairness"

- Independence
- Separation (Equalized odds)
- Sufficiency

These correspond to equalizing across groups

- Acceptance rate $\mathbb{P}\{\hat{Y} = 1\}$ of the classifier
- The error rates $\mathbb{P}\{\hat{Y} = 0 | Y = 1\}$ and $\mathbb{P}\{\hat{Y} = 1 | Y = 0\}$ of the classifier
- Outcome frequency $\mathbb{P}\{Y = 1 | R = r\}$ of a score R

Independence

- **Definition:**
  The model R satisfies *independence* if $R \perp A$.
- Applied to binary predictions $\hat{Y}$, this is equivalent to

$$\mathbb{P}\{\hat{Y} = 1 | A = a\} = \mathbb{P}\{\hat{Y} = 1 | A = b\}$$

  for all groups a, b
- That is, equalizing the acceptance rate $\mathbb{P}\{\hat{Y} = 1\}$ of the classifier across groups

Question

In which ways is this a *good* criterion? In which ways is it *bad*? E.g. consider scenarios:

- Hiring female or male software developers
- Diagnosing cancer



Separation (Equalized Odds)

- **Definition:**
  The model R satisfies *separation* if $R \perp A | Y$.
- **Intuition:** The target Y quantifies underlying *merit*
- Applied to binary predictions $\hat{Y}$, this is becomes:

$$\mathbb{P}\{\hat{Y} = 1 | Y = 1, A = a\} = \mathbb{P}\{\hat{Y} = 1 | Y = 1, A = b\}$$

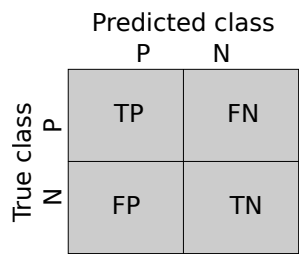
$$\mathbb{P}\{\hat{Y} = 1 | Y = 0, A = a\} = \mathbb{P}\{\hat{Y} = 1 | Y = 0, A = b\}$$

- Equalizing *true positive rate* and *false positive rate*
- Equivalent: Equalizing *false negative rate* and *true negative rate*

Question

In which ways is this a *good* or *bad* criterion? E.g.:

- Hiring female and male software developers
- Diagnosing cancer



Sufficiency

- The model R satisfies *sufficiency* if $Y \perp A | R$
- That is, for all groups a, b and all r, we have

$$\mathbb{P}\{Y = 1 | R = r, A = a\} = \mathbb{P}\{Y = 1 | R = r, A = b\}$$

- For binary classifiers $f(X) = \hat{Y} \in \{0, 1\}$, this indicates equal positive and negative predictive values

$$\mathbb{P}\{Y = 1 | \hat{Y} = 1, A = a\} = \mathbb{P}\{Y = 1 | \hat{Y} = 1, A = b\}$$

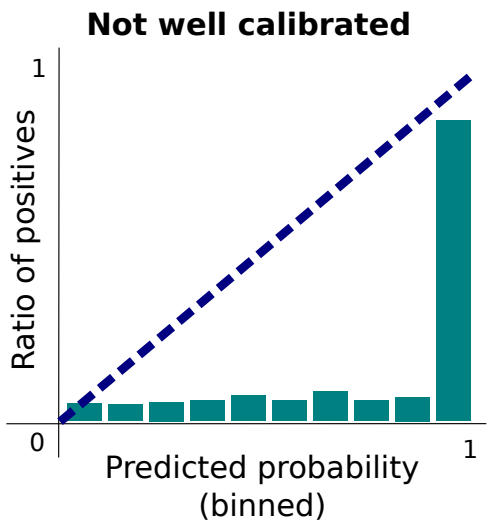
$$\mathbb{P}\{Y = 0 | \hat{Y} = 0, A = a\} = \mathbb{P}\{Y = 0 | \hat{Y} = 0, A = b\}$$

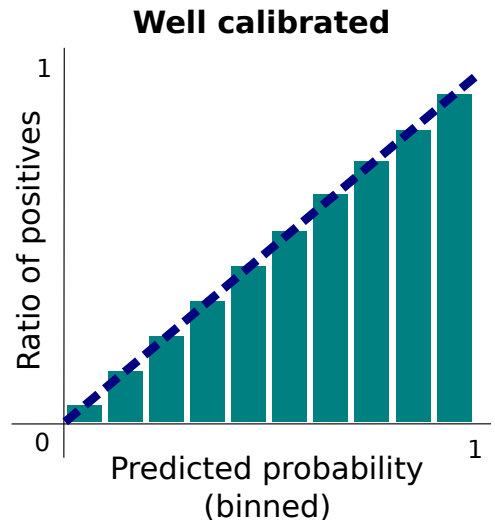
Question

In which ways is this a *good* criterion? In which ways is it *bad*?

How sufficiency relates to calibration

- Sufficiency – for probabilistic classifiers – corresponds to equalizing the outcome frequency $\mathbb{P}\{Y = 1|R = r\}$ of the score R across groups A for all scores $r \in R$.
- This is closely related to *calibration*.



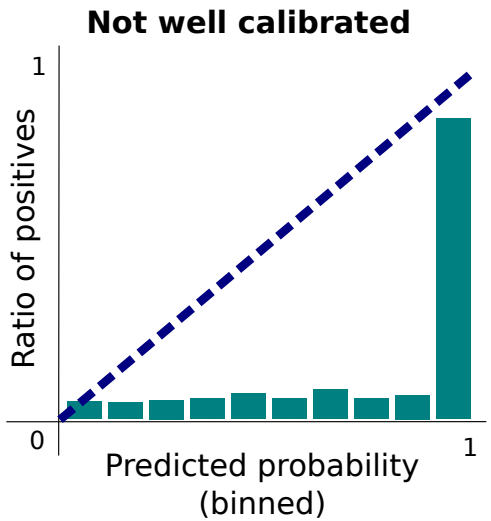


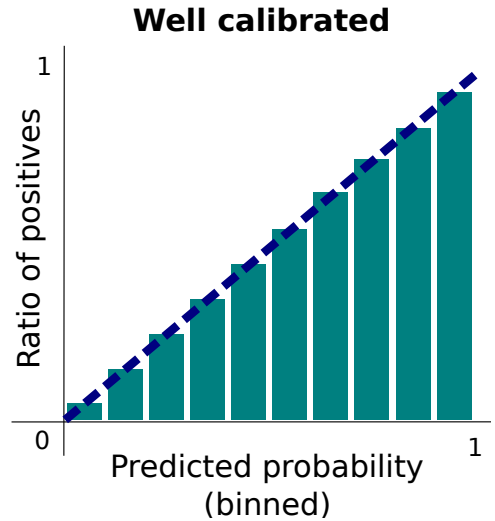
How sufficiency relates to calibration

- Sufficiency – for probabilistic classifiers – corresponds to equalizing the outcome frequency $\mathbb{P}\{Y = 1|R = r\}$ of the score R across groups A for all scores $r \in R$.
- This is closely related to *calibration*.
- A model R is *calibrated* with respect to the target Y if for all scores $r \in [0, 1]$, we have

$$\mathbb{P}\{Y = 1|R = r\} = r.$$

- Equivalently: on a population level, the predicted foreground probability equals the empirical one.





How sufficiency relates to calibration

Why is this interesting?

- Sufficiency:
  $\mathbb{P}\{Y = 1 | R = r, A = a\} = \mathbb{P}\{Y = 1 | R = r, A = b\}$
- Calibration: $\mathbb{P}\{Y = 1 | R = r\} = r$

Theorem

- *Calibration by group implies sufficiency*
- *If a model R satisfies sufficiency, then there exists a calibrating function $l : [0, 1] \to [0, 1]$ so that $l(R)$ satisfies calibration by group.*

How sufficiency relates to calibration

Why is this interesting?

- Sufficiency:
  $\mathbb{P}\{Y = 1 | R = r, A = a\} = \mathbb{P}\{Y = 1 | R = r, A = b\}$
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Theorem

- *Calibration by group implies sufficiency*
- *If a model R satisfies sufficiency, then there exists a calibrating function l: $[0, 1] \to [0, 1]$ so that $l(R)$ satisfies calibration by group.*

Calibration is well studied and we have off-the-shelf tools to calibrate models. This means we can obtain sufficiency through calibration.

Mutually exclusive: Independence ($R \perp A$) vs Sufficiency ($Y \perp A|R$)

Assumption: A and Y are not independent

Claim: Then sufficiency and independence cannot both hold

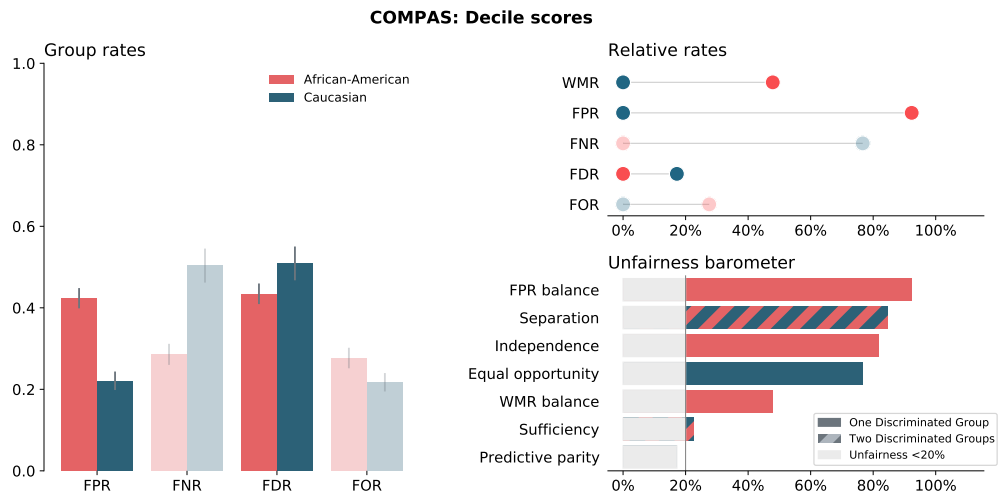


Figure: Figure from MSc thesis of CF Damgaard and E Zinck, 2022

Mutually exclusive: Independence ($R \perp A$) vs Separation ($R \perp A|Y$)

Assumption: Y is binary, A is not independent of Y, R is not independent of Y.
Claim: Then, independence and separation cannot both hold

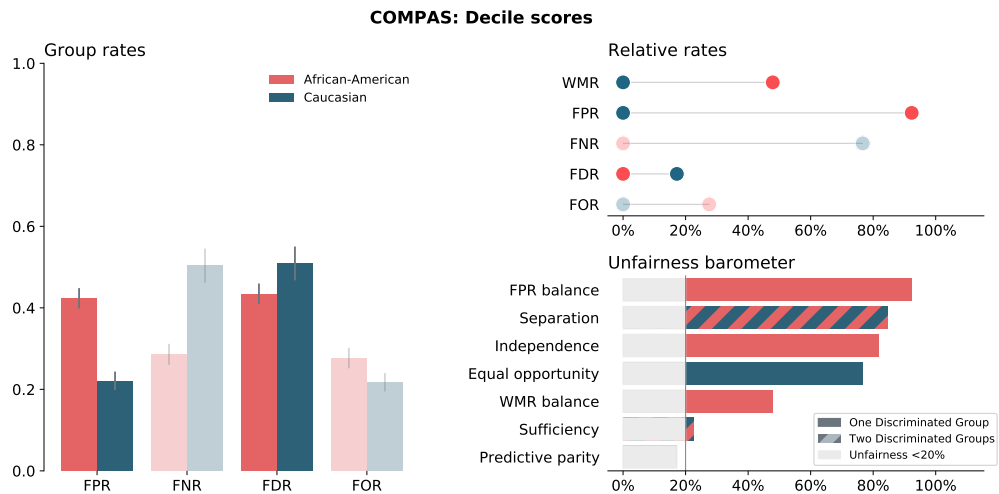


Figure: Figure from MSc thesis of CF Damgaard and E Zinck, 2022

Mutually exclusive: Sufficiency ($Y \perp A|R$) vs Separation ($R \perp A|Y$)

Assumption: All events in the joint distribution of (A, R, Y) have positive probability, and A is independent of Y. Then, separation and sufficiency cannot both hold.

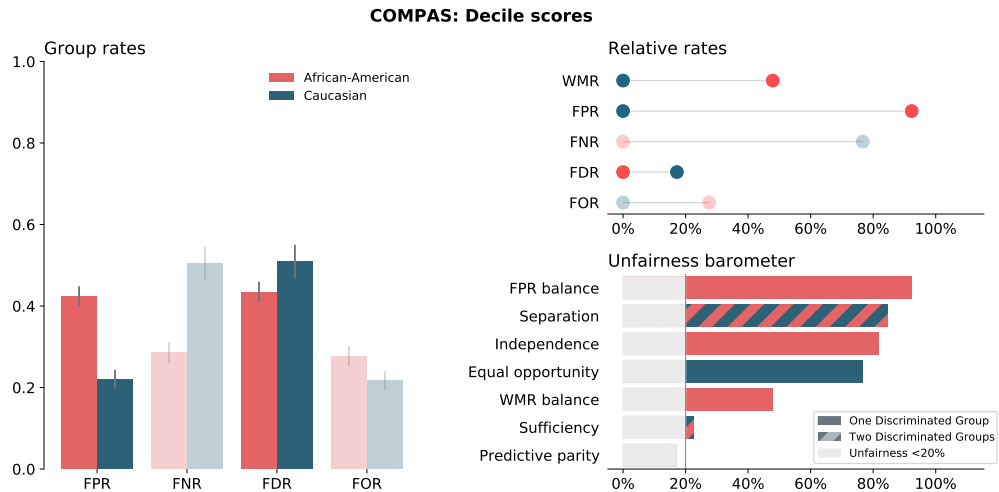


Figure: Figure from MSc thesis of CF Damgaard and E Zinck, 2022

Next:
Bias diagnostics: From fairness criterion to quantification

Diagnostics based on algorithmic fairness criteria

Recall: The three categories of "fairness" correspond to equalizing across groups

- **Independence:** Acceptance rate $\mathbb{P}\{\hat{Y} = 1\}$ of the classifier
- **Separation:** The error rates $\mathbb{P}\{\hat{Y} = 0 | Y = 1\}$ and $\mathbb{P}\{\hat{Y} = 1 | Y = 0\}$ of the classifier
- **Sufficiency:** Outcome frequency $\mathbb{P}\{Y = 1 | R = r\}$ of a score R

Diagnosing bias is easy! A simple diagnostic of "unfairness" can be derived by looking at either difference or fractions of the associated rates or frequencies between groups.

Tools

A number of software tools are freely available, of different quality and usability. Let's see an example from the Fairlearn package, which offers easy bias diagnostics:

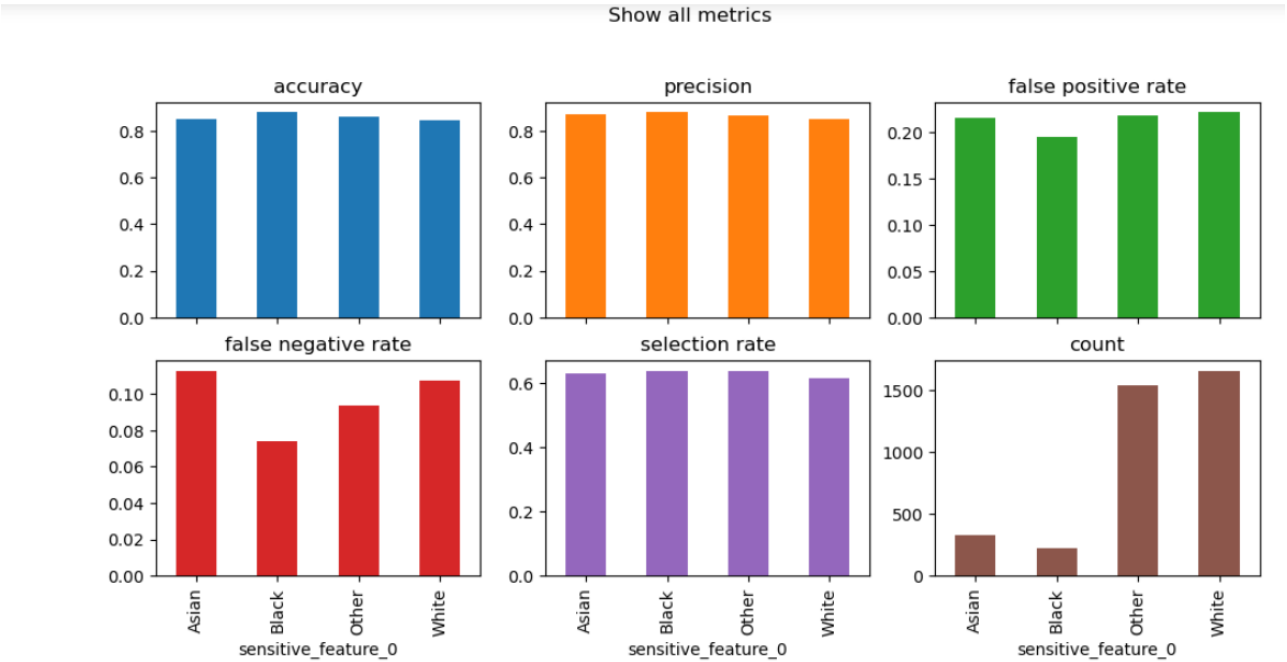


Figure: Fairness diagnostics with Fairlearn

Irvin, Jeremy, et al. AAAI'19

Bird Sarah, et al. Microsoft, Tech. Rep. MSR-TR-2020-32 (2020).

Summary

By now, you should:

- Be familiar with different ways in which demographic bias can become embedded in machine learning models
- Be familiar with the three most common fairness criteria, and variants thereof
- Know how to turn fairness criteria into diagnostic criteria for algorithmic bias
- Be familiar with incompatibility results for fairness criteria
- Be ready to start playing with algorithmic bias detection and mitigation with FairLearn